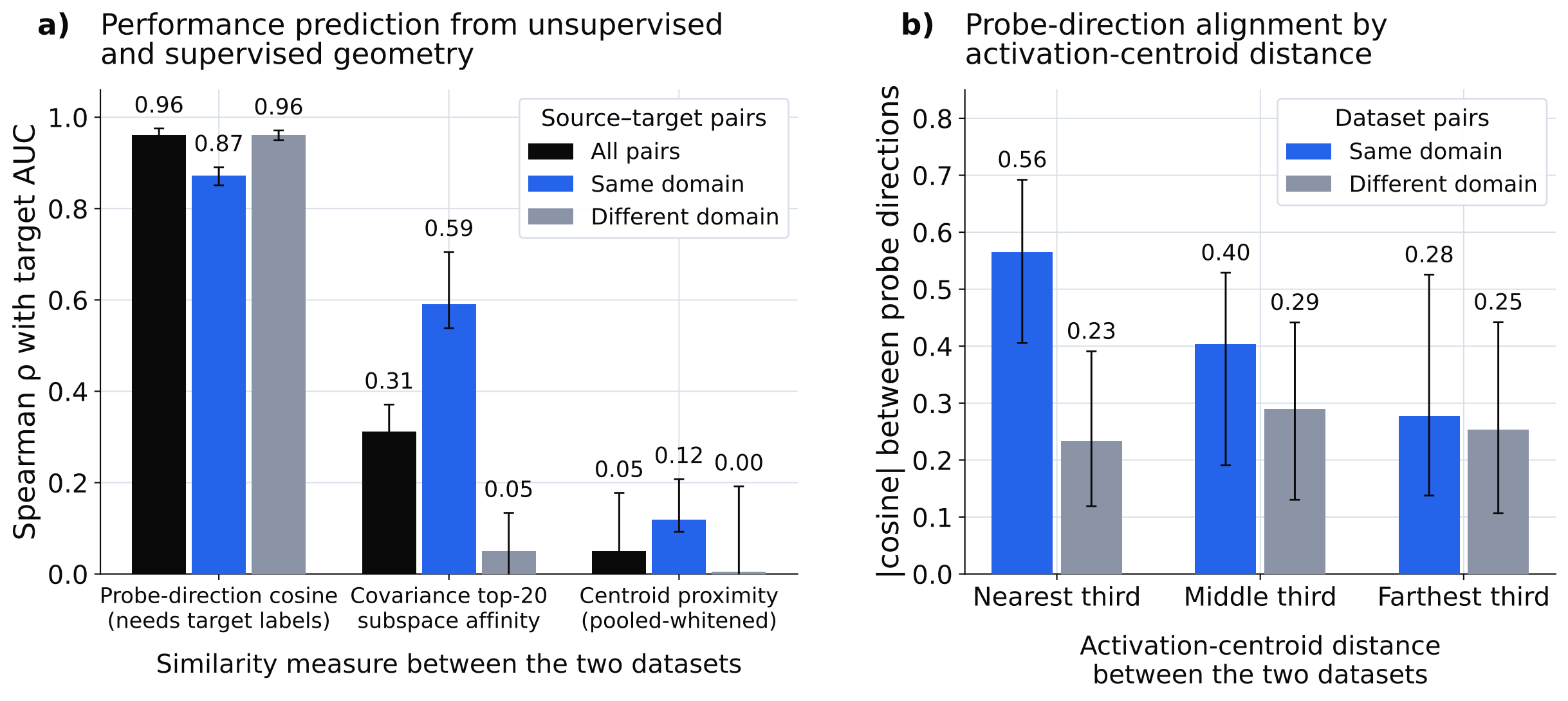
a)
Performance prediction from unsupervised and supervised geometry
Spearman ρ with target AUC
1.0
0.8
0.6
0.4
0.2
0.0
0.96
0.87
0.96
0.31
0.59
0.05
0.05
0.12
0.00
Source–target pairs
All pairs
Same domain
Different domain
Probe-direction cosine (needs target labels)
Covariance top-20 subspace affinity
Centroid proximity (pooled-whitened)
Similarity measure between the two datasets
b)
Probe-direction alignment by activation-centroid distance
|cosine| between probe directions
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.56
0.23
0.40
0.29
0.28
0.25
Dataset pairs
Same domain
Different domain
Nearest third
Middle third
Farthest third
Activation-centroid distance between the two datasets